Anton Lee

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Education

University of Waterloo Sep. 2024 – April 2029
Bachelor of Applied Science in Computer Engineering *Waterloo, ON*

Skills

Languages: Python, Java, C/C++, C#, GoLang, TypeScript/JavaScript, HTML/CSS
Frameworks: React, Next.JS, Node.JS, Express, Flask, JQuery, RestAPI, Tailwind, Three.js, .NET

Developer Tools: Git, Jira, Figma, Excel, MongoDB, AWS, Google Cloud (GCP), Azure, PostgresSQL/MySQL, CI/CD
Libraries: pandas, NumPy, OpenCV, Tensorflow, PyTorch, OpenGL, Qt, VTK/ITK, DCMTK

Experience

Software Engineer and Researcher | *SickKids* *January 2026 – April 2026 | Toronto, ON*
- Implemented interactive 2D/3D visualization and segmentation using C++/Qt/VTK/ITK with DICOM support via DCMTK.
- Reduced visualization latency on CT volumes by 10x through GPU-accelerated rendering, threaded processing, and caching.

Software Development Tester | *Lumeto Inc.* *May 2025 – August 2025 | Toronto, ON*
- Designed and performed stress and performance tests under varied configurations, validating product stability and preventing environment-specific failures prior to release.
- Found and documented bugs in Jira, reducing resolution time through detailed reproduction steps and severity assessments.
- Authored and maintained streamlined test plans, cases, and regression suites, contributing to a structured QA pipeline.
- Collaborated with development team to catch issues and validate fixes, preventing regressions from reaching production.

Startup Co-Founder & COO | *Flair Studio Inc.* *June 2024 – Present | Toronto, ON*
- Co-founded an AI-powered shopping aggregator and social platform for online fashion discovery, featuring autonomous web scraping, custom recommendation algorithms, and interactive user closets.
- Designed a scalable backend with Flask, AWS Lambda, MongoDB to process user data and personalized search in real-time.
- Developed an interactive frontend prototype with Next.js/React, Framer Motion, and GSAP to design an addictive scrollable UI for an immersive user experience.
- Fine tuned an Azure OpenAI model to optimize a fashion chatbot, and Implemented an AI-powered recommendation systems using collaborative filtering, content-based filtering, reinforcement learning.

Projects

Lattice | *Finalist / Winner @ Hack the North 2025* *September 2025*
C++, C#, .NET, OpenCV, Distributed Computing, LiDAR
- Built a real-time 3D scanning/reconstruction system that fuses multi-camera and LiDAR inputs into a unified volumetric model.
- Designed a distributed client–server architecture with per-sensor data (depth, colour) streaming, improving synchronization.
- Implemented point cloud filtering and merging algorithms to remove noise, providing more accurate 3D reconstructions.
- Developed a two-step calibration pipeline, implementing Procrustes Orthogonal problem to generate relative matrix transformations with OpenCV marker detection and ICP refinement for precise multi-sensor alignment.
- Architected a lock-free rolling zero-copy ring buffer, maintaining a bounded in-memory window of sensor data streams to enable instant temporal rewind and deterministic 6DoF playback of volumetric scenes without disk writes.

DataBot | *Finalist @ LAHacks 2025* *April 2025*
Go, Typescript, Next.js, Google Gemini
- Designed an AI-driven platform that optimizes slow queries, storage bloat, and unused data for PostgreSQL, MySQL, MongoDB.
- Integrated Gemini 2.5 for real-time SQL rewrite suggestions, reducing query latency by 40% and CPU usage by 25%.
- Created a modular Go backend and streaming pipeline able to process 1M+ events a day, ensuring scalablility.
- Built a Next.js/React dashboard showing CO_2 and kWh savings; "Green Score" reflects database efficiency and sustainability.

Hello World | *1st Place @ UTS BluesHacks 2024, Student Hacks* *January 2024*
Python, JavaScript, OpenCV, OpenPose, Pyinput, MediaPipe, Google Cloud API
- Built an animated video-to-text/audio ASL interpreter, improving communication accuracy by 35% for the hearing-impaired.
- Created a frame compression and processing algorithm, reducing average processing time to 0.8 seconds, achieving 95% translation accuracy from ASL Video to English.
- Implemented TensorFlow, MediaPipe, OpenPose, HumanGAN to extract skeletal position and create fluid signing animations.

Verbate | *Best use of AI @ IEEE MakeUC 2023* *November 2023*
Python, Node.js, React.js, Next.js, GPT-4.0, AWS
- Integrated ChatGPT-4.0 and AWS Polly, creating a GenAI-powered interview preparation app featuring voice input and text-to-speech capabilities, enhancing user interaction and engagement.
- Utilized React.js, and Radix UI to build a responsive front-end and efficient back-end infrastructure, enabling a seamless UX.